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SUBTYPE DISCOVERY IN PCOS PATIENTS: AN UNSUPERVISED LEARNING AND EXPLAINABILITY APPROACH

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Abstract
Polycystic Ovary Syndrome (PCOS) is a complex endocrine disorder that affects women in diverse ways, leading to considerable variation in hormonal, metabolic, and reproductive characteristics. Due to this heterogeneity, conventional diagnostic approaches often fail to capture underlying differences among patients. This study aims to explore whether meaningful PCOS subtypes can be identified using unsupervised machine learning techniques applied to clinical biomarker data. A structured preprocessing pipeline was implemented, including correction of inconsistent values, median-based imputation for missing data, and feature standardization. Seventeen clinically relevant biomarkers representing hormonal, metabolic, and ovarian characteristics were selected. Principal Component Analysis (PCA) was employed to reduce dimensionality while retaining essential data variation. Clustering was performed using K-Means, Gaussian Mixture Models, and Agglomerative Clustering on data from 177 confirmed PCOS patients. Silhouette scores were calculated to evaluate clustering quality. Among the tested methods, K-Means demonstrated superior separation performance. Subsequent analysis revealed two dominant patient subtypes: Lean PCOS and Metabolic/Insulin-Resistant PCOS. Explainability analysis further identified key biomarkers contributing to subtype differentiation. The findings support the use of data-driven approaches for understanding PCOS heterogeneity and provide a foundation for future personalized clinical strategies.
Keywords: Polycystic Ovary Syndrome; Unsupervised Learning; Clustering Analysis; Dimensionality Reduction; Biomarker Profiling.

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1. Introduction

Polycystic Ovary Syndrome (PCOS) is one of the most frequently diagnosed endocrine disorders among women of reproductive age. It is associated with irregular menstrual cycles, hormonal imbalance, ovarian dysfunction, insulin resistance, and metabolic disturbances. Although standardized diagnostic criteria exist, PCOS does not present uniformly across patients. Some individuals exhibit predominantly reproductive symptoms, whereas others show significant metabolic complications. This diversity suggests that PCOS represents a spectrum rather than a single homogeneous condition.

Traditional classification systems rely primarily on symptom-based criteria. While these systems are clinically practical, they may not reflect deeper biological variation among patients. Consequently, individuals with distinct metabolic profiles may be grouped under the same diagnostic label, potentially limiting personalized treatment strategies. Understanding heterogeneity within PCOS is therefore critical for improving patient management.

With increasing availability of clinical datasets, machine learning methods offer new opportunities to identify hidden patterns within complex medical data. Unsupervised learning, in particular, enables grouping of patients based on similarity across multiple variables without predefined labels. Such approaches are well suited for subtype discovery in heterogeneous conditions. This study applies unsupervised clustering techniques to identify potential PCOS subtypes using multidimensional biomarker data.

2. Methodology

The overall workflow adopted in this study is illustrated in **Figure 1**, which outlines the sequential steps from data preparation to subtype identification and interpretation.

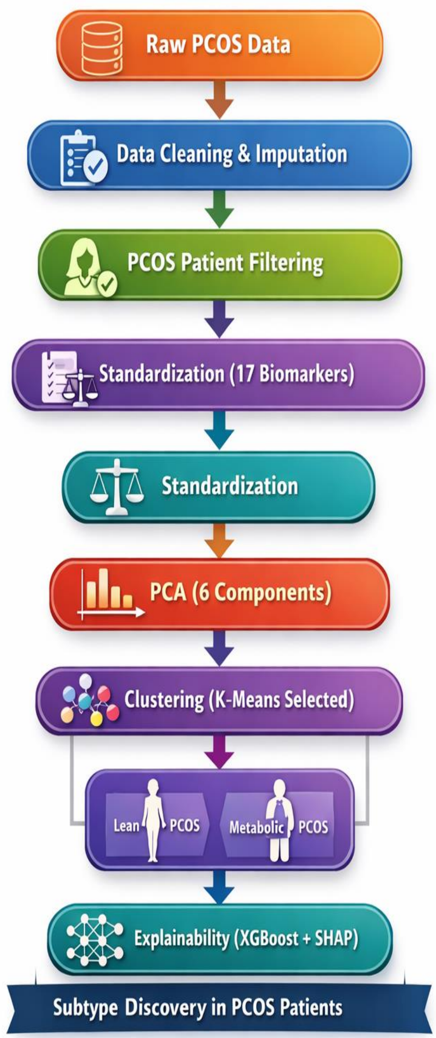

Figure 1. Workflow illustrating data cleaning, feature selection, standardization, PCA, clustering valuation, subtype identification, and explainability analysis.

2.1 Data Preparation

The dataset consisted of clinical and biochemical measurements collected from women diagnosed with PCOS. Initial inspection revealed missing values and inconsistencies, particularly in hormonal measurements such as Anti-Müllerian Hormone (AMH). Median imputation was applied to handle missing entries while maintaining dataset completeness.

Only confirmed PCOS cases were retained, resulting in a final dataset of 177 patients. This ensured that clustering analysis focused exclusively on the relevant population.

2.2 Feature Selection and Standardization

Seventeen biomarkers were selected based on clinical importance, covering hormonal, metabolic, menstrual, cardiovascular, and ovarian domains. Because these features were measured on different numerical scales, standardization was performed to ensure equal contribution during clustering.

2.3 Dimensionality Reduction Using PCA

High-dimensional datasets may contain redundant information and noise. Principal Component Analysis (PCA) was applied to transform the original variables into principal components that retained most of the variance. Six principal components were selected based on cumulative variance explained. This step simplified the dataset while preserving essential patterns.

2.4 Clustering Algorithms

Three unsupervised clustering algorithms were evaluated:

- K-Means
- Gaussian Mixture Models
- Agglomerative Clustering

Silhouette scores were calculated to assess cluster compactness and separation. The algorithm with the highest silhouette score was selected for subtype identification.

3. Results and Discussion

The presentation of results follows the same order as the analytical workflow. Quantitative evaluation of clustering performance is presented first, followed by visual cluster representation and subtype interpretation. This alignment ensures logical progression from methodological justification to interpretation of findings.

3.1 Clustering Performance Evaluation

Silhouette score comparison indicated that K-Means achieved stronger separation compared to Gaussian Mixture Models and Agglomerative Clustering. This suggests better-defined clusters and improved internal consistency.

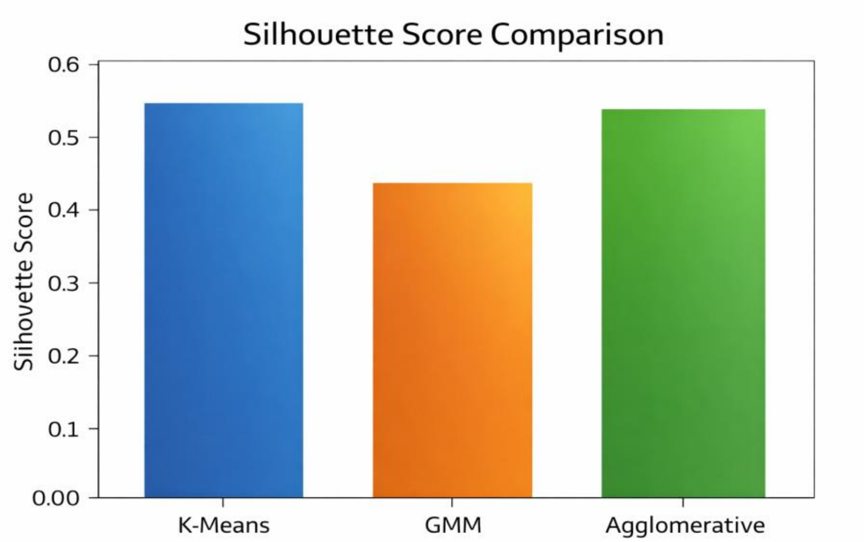

Figure 2. Comparison of silhouette scores across clustering algorithms, demonstrating superior performance of K-Means.

Based on this evaluation, K-Means was selected for further analysis and subtype identification.

3.2 Visualization of Clusters

To better understand the structure of the clusters, the PCA-transformed data were projected onto the first two principal components. The resulting scatter plot provided a clear visual representation of patient group separation within reduced-dimensional space.

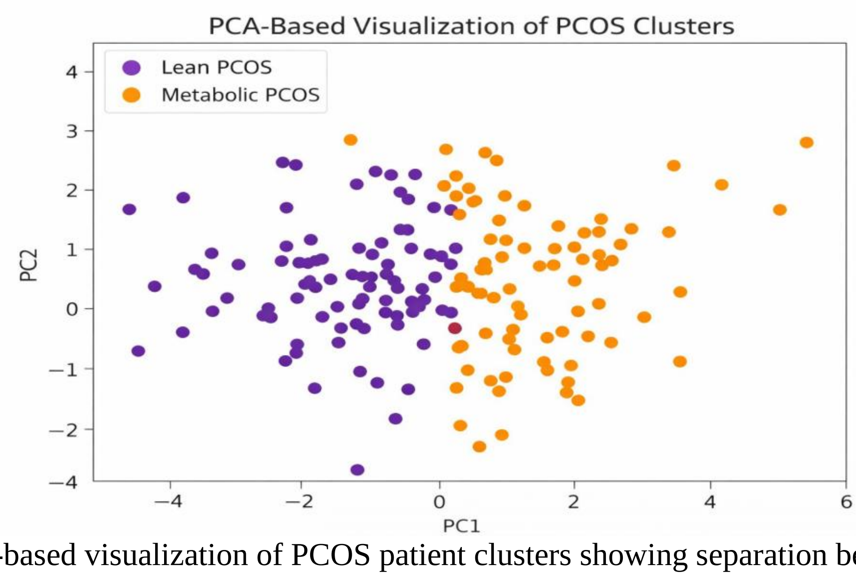

Figure 3. PCA-based visualization of PCOS patient clusters showing separation between Lean PCOS and Metabolic PCOS subtypes.

The visual separation supports the quantitative silhouette score findings and indicates the presence of distinct subtype patterns.

3.3 Identified PCOS Subtypes

Further examination of cluster centroids and biomarker distributions led to identification of two dominant subtypes:

Subtype 1: Lean PCOS (138 patients)

Patients in this group showed relatively normal BMI and metabolic markers while still presenting PCOS symptoms. This subtype emphasizes that PCOS is not exclusively associated with obesity or severe metabolic imbalance.

Subtype 2: Metabolic / Insulin-Resistant PCOS (39 patients)

This group exhibited higher BMI and metabolic indicators suggestive of insulin resistance. These patients may have increased risk of long-term metabolic complications.

3.4 Interpretability

Explainability analysis identified BMI, AMH, weight, and follicle count as influential variables contributing to subtype differentiation. These findings align with known clinical characteristics and enhance confidence in the biological relevance of the identified clusters.

4. Implications of Findings

The identification of two distinct PCOS subtypes highlights the importance of personalized evaluation. Lean PCOS patients may require hormonal management strategies, whereas metabolically affected patients may benefit from targeted metabolic interventions. Data-driven subtype discovery can therefore support improved clinical decision-making.

5. Limitations

Although meaningful subtypes were identified, the study has certain limitations. The dataset size was moderate, and validation using independent datasets was not performed. Additionally, longitudinal follow-up data were not available to assess subtype stability over time. Future research should address these limitations to strengthen generalizability.

6. Future Scope

Future studies may expand this research by incorporating larger datasets, integrating additional biomarkers, and applying longitudinal tracking. Combining clustering results with supervised predictive models may further enhance clinical utility.

7. Conclusion

This study demonstrates that unsupervised machine learning can uncover meaningful heterogeneity within PCOS populations. Through structured preprocessing, dimensionality reduction, clustering evaluation, and interpretability analysis, two clinically relevant subtypes were identified. These findings reinforce the importance of data-driven approaches in understanding complex endocrine disorders and provide a foundation for future personalized healthcare strategies.

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